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DEPARTMENT OF DATA SCIENCE
III CSD Industry Oriented Mini Projects A.Y. 2025-2026
Team - 16

AI POWERED POTHOLE DETECTION.

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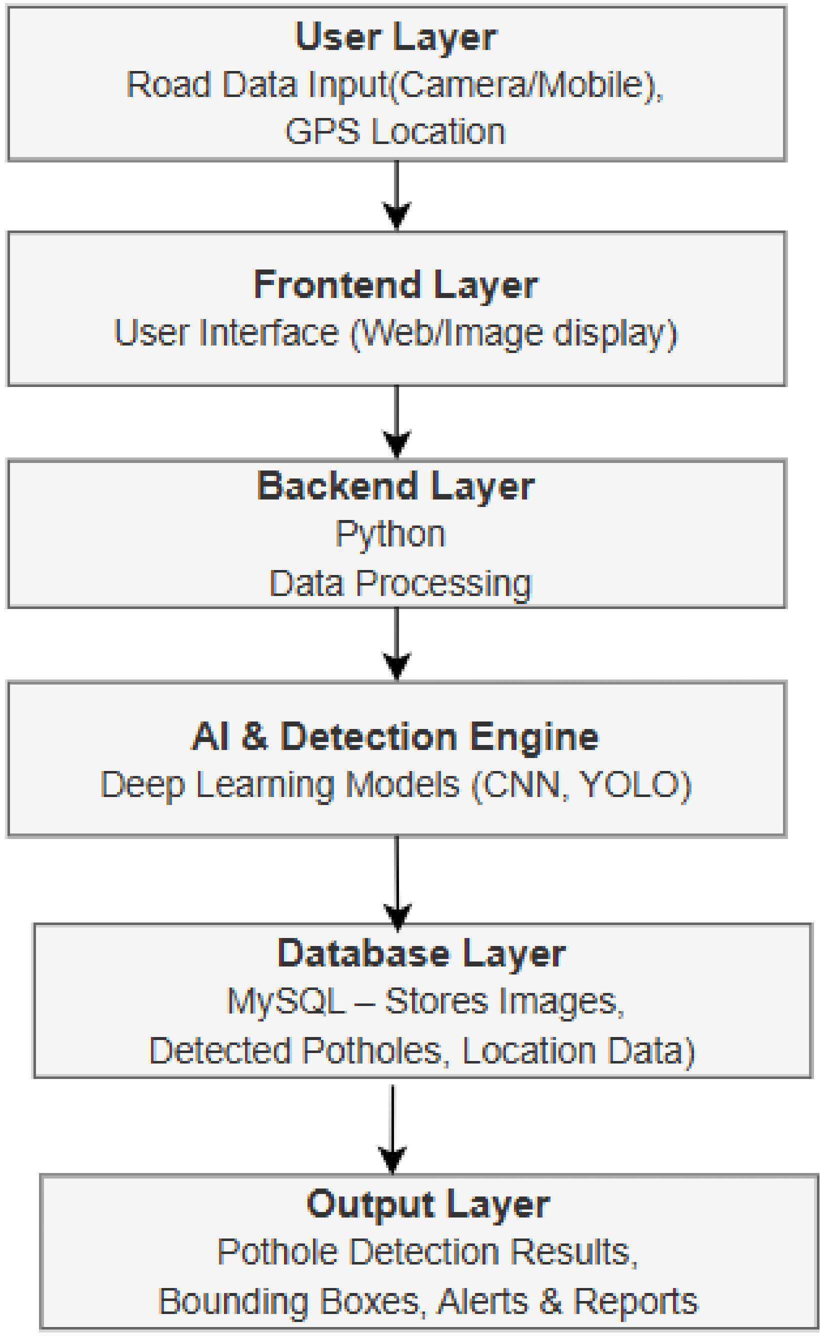
Abstract

Road infrastructure plays a crucial role in economic development and public safety. However, potholes and road surface damages are major challenges in urban and rural transportation systems. These defects often lead to traffic accidents, vehicle damage, increased fuel consumption, and traffic congesti on. Traditional road inspection methods are manual, time-consuming, costly, and inefficient. To address this issue, this project proposes an AI-based Pothole Detection System using Deep Learning techniques, specifically Convolutional Neural Networks (CNN) and YOLO (You Only Look Once) object detection models.

Literature Survey Summary Table

S.No	IEEE Paper Title	Authors	Methodology	Advantages	Disadvantages
1	AI-Based Pothole Detection System Using Deep Learning Techniques.	R. Sharma, P. Kumar, S. Verma, A. Singh	CNN and YOLO- base object detection using road images and videos	• High detection accuracy • Real-time detection • Reduces manual inspection.	• Requires large dataset • Performance affected by lighting conditions
2	Real-Time Road Damage Detection Using YOLO and Image Processing Techniques.	M. Patel, K. Reddy, S. Gupta, N. Verma.	Yolo based real time detection with image processing.	• Fast processing speed • Suitable for real-time systems • Easy implementation	• High computational cost • Needs training time
3	Automated Road Crack and Pothole Detection Using Deep Learning	S. Li, Y. Zhang, H. Wang, X. Chen	CNN-based automated detection using image datasets	• High accuracy • Robust feature extraction • Works on large datasets	• Requires hardware setup • Higher implementation cost
4	Smart road surface monitoring system.	A. Kumar, V. Singh, R. Mishra, P. Sharma	Combination of IoT sensors and computer vision techniques	• Real-time monitoring • Better data collection • Scalable system	• Requires hardware setup • Higher implementation cost
5	Real time pothole detection and Reporting System.	J. Brown, L. Davis, M. Wilson, E. Clark	ML based detection with GPS.	• Location-based detection • Automated reporting • Improves maintenance planning	• Depends on network connectivity • Accuracy varies with environment

System Architecture



Algorithm/Methodology

Image Capture & Cleaning
Camera captures road images → Images are resized → Unwanted areas (sky, trees) are cropped.

Pothole Detection (AI Model)
Input image passed to YOLO model → Model detects potholes → Bounding boxes are created → Number of potholes is counted

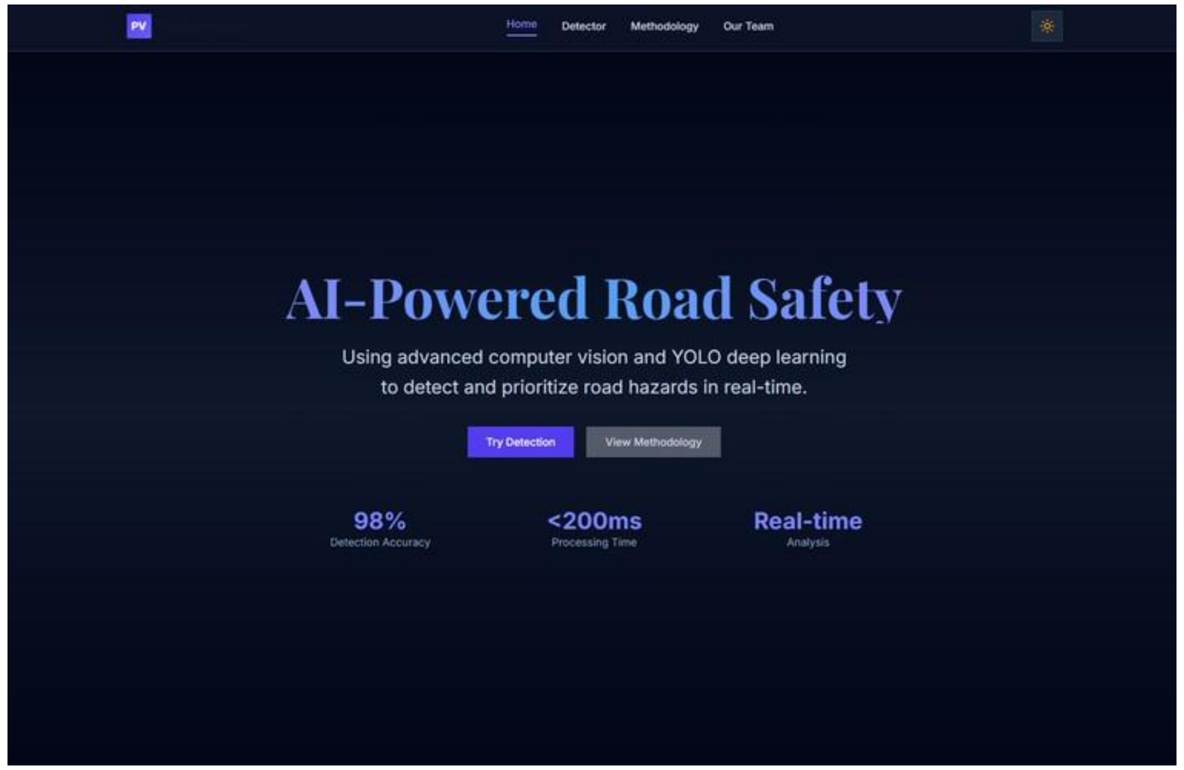
Location & Data Capture
GPS location (latitude, longitude) is collected →data checks how severe the pothole is→ All data is stored with the image.

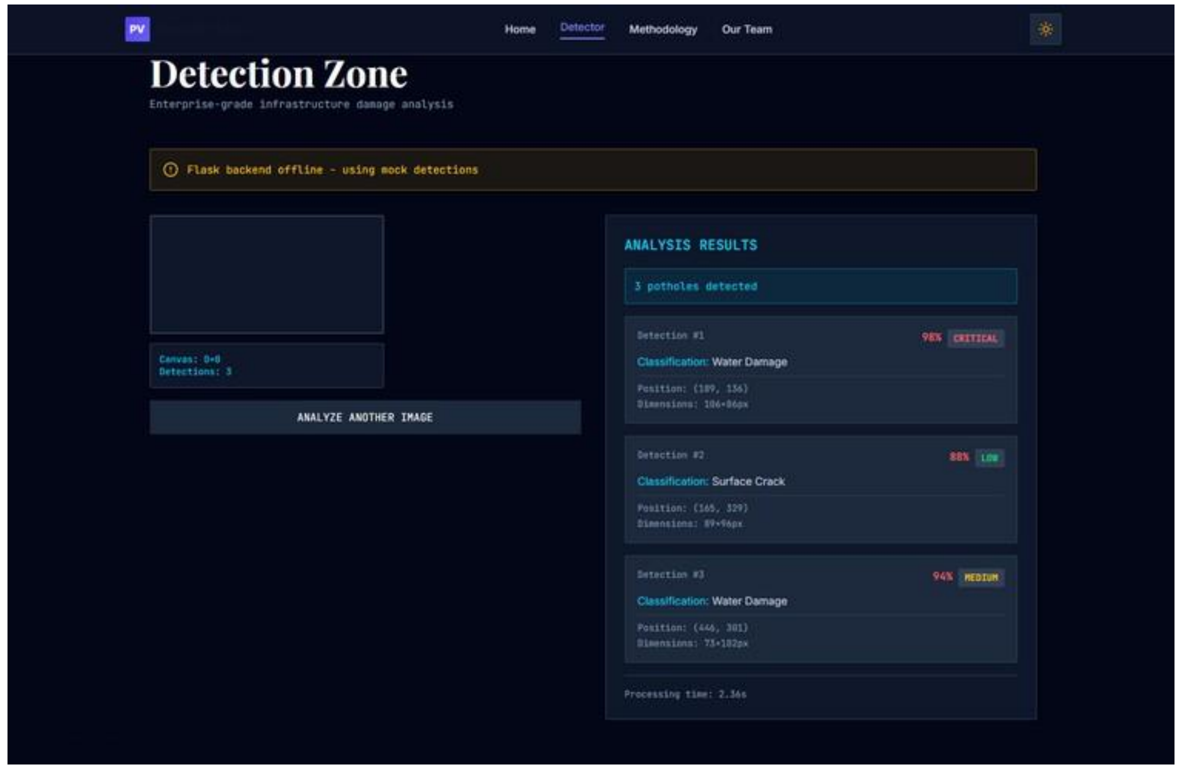
Reporting & Sending Data
Detected data sent to server → Real-time reporting is triggered → Data stored in central system.

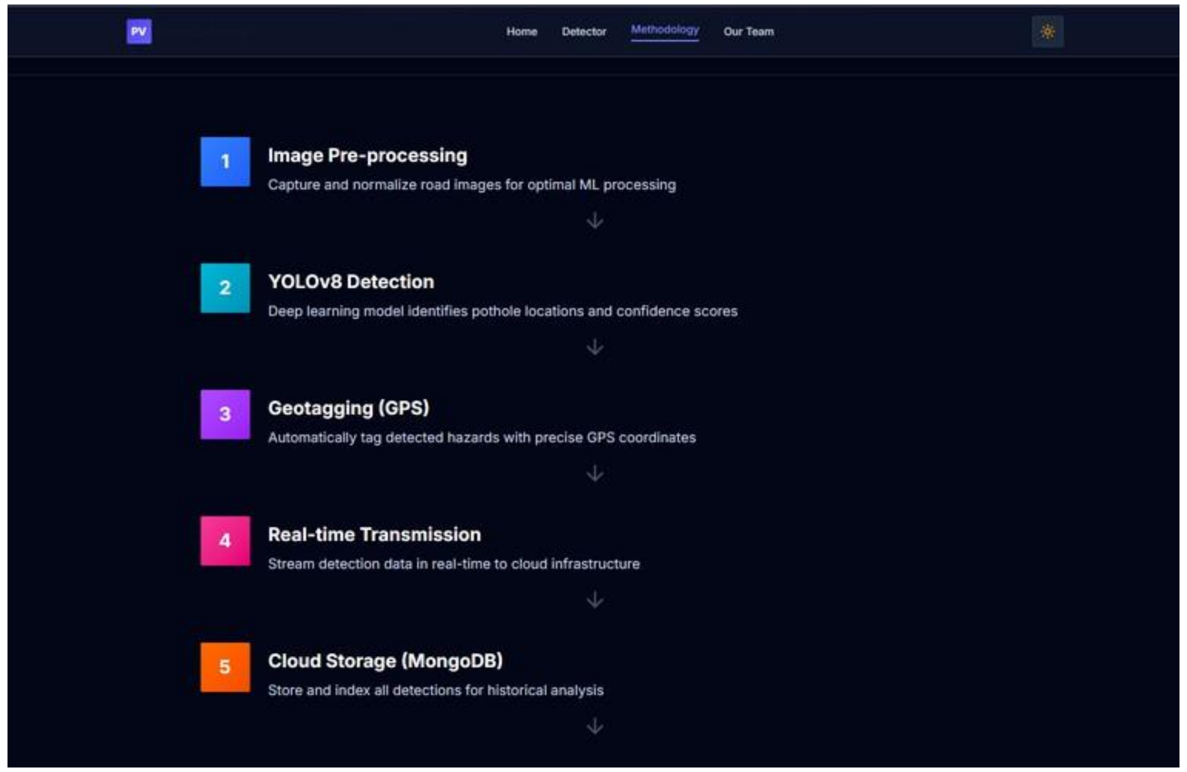
Data Storage & Priority
Data stored in database (MySQL) → Potholes prioritized based on severity → Historical data maintained.

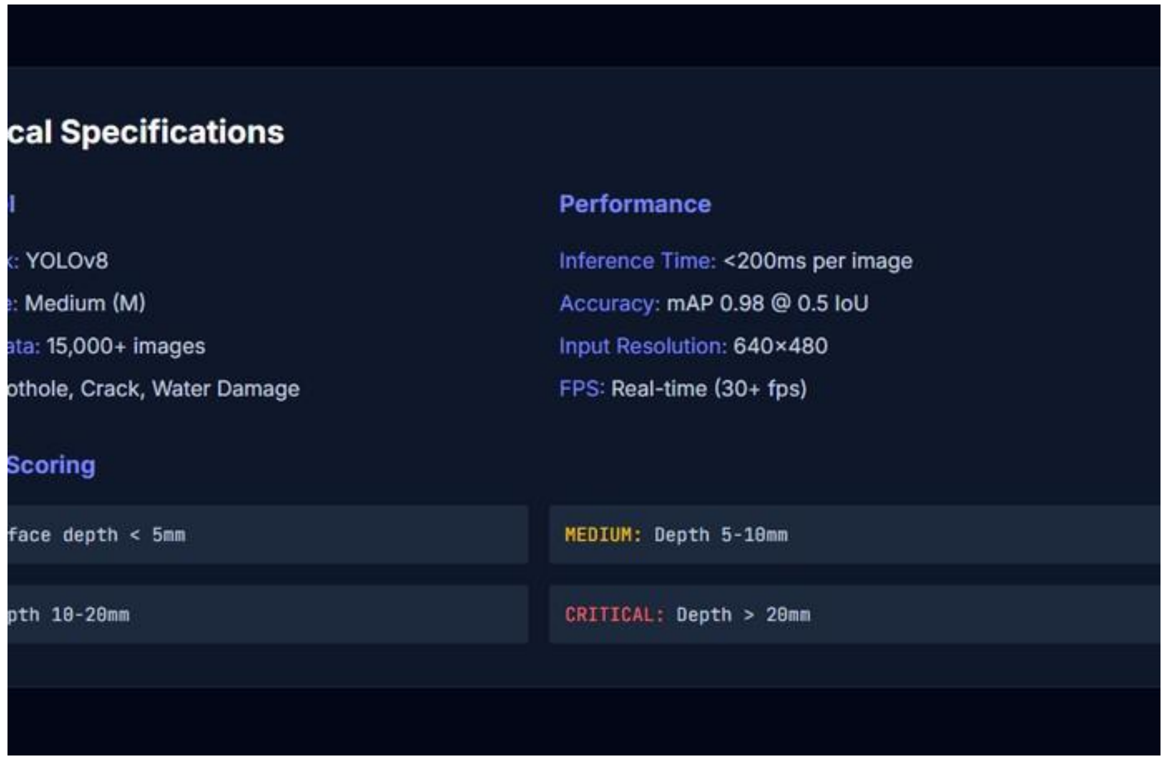
Visualization & Alerts
Data plotted on map → Dashboard updated → Alerts sent to authorities.

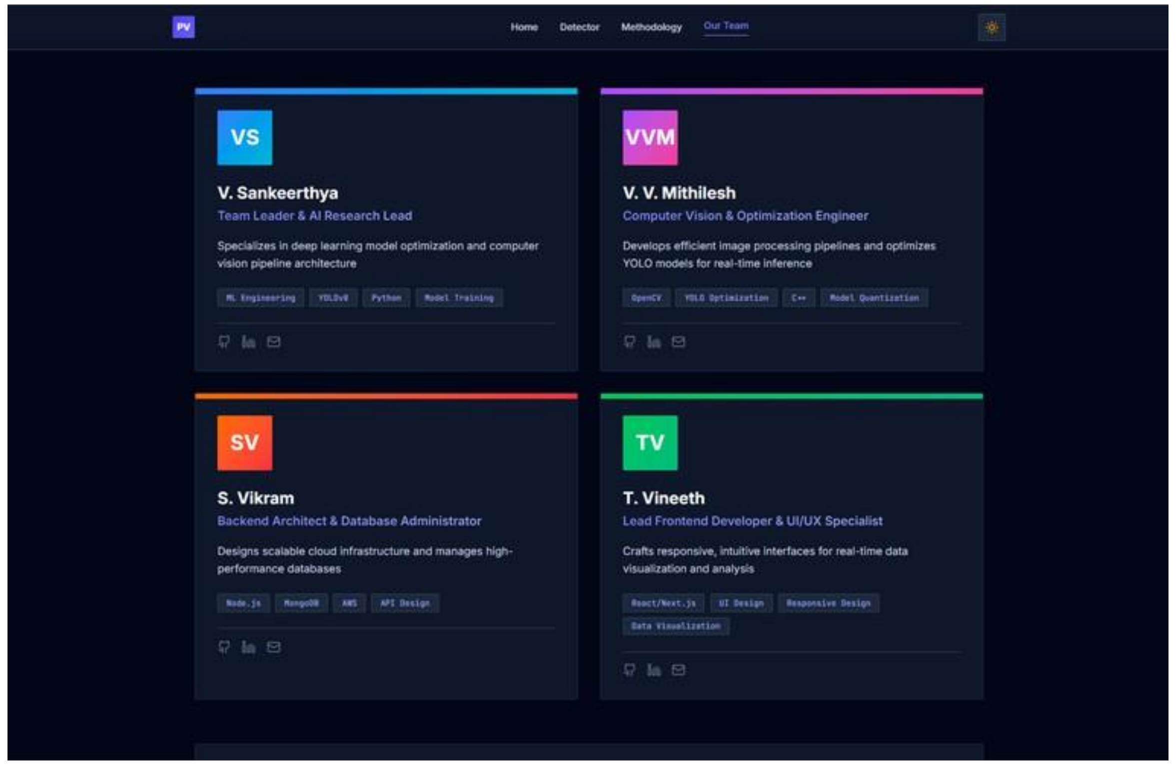
Results


AI-Powered Road Safety
Using advanced computer vision and YOLO deep learning
to detect and prioritize road hazards in real-time.
Try Detection
View Methodology
98%
Detection Accuracy
<200ms
Processing Time
Real-time
Analysis


Detection Zone
Enterprise-grade infrastructure damage analysis
Flask backend offline - using mock detections
ANALYZE ANOTHER IMAGE
ANALYSIS RESULTS
3 potholes detected
Classification: Water Damage
Classification: Surface Crack
Classification: Water Damage


1
Image Pre-processing
Capture and normalize road images for optimal ML processing
2
YOLOv8 Detection
Deep learning model identifies pothole locations and confidence scores
3
Geotagging (GPS)
Automatically tag detected hazards with precise GPS coordinates
4
Real-time Transmission
Stream detection data in real-time to cloud infrastructure
5
Cloud Storage (MongoDB)
Store and index all detections for historical analysis


cal Specifications
Performance
: YOLOv8
: Medium (M)
ata: 15,000+ images
othole, Crack, Water Damage
Inference Time: <200ms per image
Accuracy: mAP 0.98 @ 0.5 IoU
Input Resolution: 640×480
FPS: Real-time (30+ fps)
Scoring
face depth < 5mm
MEDIUM: Depth 5-10mm
pth 10-20mm
CRITICAL: Depth > 20mm


VS
V. Sankeerthya
Team Leader & AI Research Lead
Specializes in deep learning model optimization and computer vision pipeline architecture
VVM
V. V. Mithilesh
Computer Vision & Optimization Engineer
Develops efficient image processing pipelines and optimizes YOLO models for real-time inference
SV
S. Vikram
Backend Architect & Database Administrator
Designs scalable cloud infrastructure and manages high-performance databases
TV
T. Vineeth
Lead Frontend Developer & UI/UX Specialist
Crafts responsive, intuitive interfaces for real-time data visualization and analysis

Conclusion

The AI-Powered Pothole Detection System presents an efficient and intelligent solution for monitoring and maintaining road infrastructure. By leveraging advanced deep learning techniques such as CNN and YOLO, the system is capable of accurately detecting potholes from images and video data in real time. The integration of preprocessing, detection, location tracking, and reporting modules ensures a complete end-to-end automated workflow.

The system reduces the dependency on manual inspection methods, thereby saving time, cost, and human effort. Additionally, the incorporation of GPS-based mapping and cloud storage enables effective tracking, reporting, and analysis of road conditions. The modular architecture, including frontend, backend, and database components, ensures scalability, flexibility, and ease of deployment in real-world environments.